

Establishing reaction networks in the 16-electron sulfur reduction reaction

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The sulfur reduction reaction (SRR) plays a central role in high-capacity lithium sulfur (Li-S) batteries. The SRR involves an intricate, 16-electron conversion process featuring multiple lithium polysulfide intermediates and reaction branches^{1–3}. Establishing the complex reaction network is essential for rational tailoring of the SRR for improved Li-S batteries, but represents a daunting challenge^{4–6}. Herein we systematically investigate the electrocatalytic SRR to decipher its network using the nitrogen, sulfur, dual-doped holey graphene framework as a model electrode to understand the role of electrocatalysts in acceleration of conversion kinetics. Combining cyclic voltammetry, in situ Raman spectroscopy and density functional theory calculations, we identify and directly profile the key intermediates (S_8 , Li_2S_8 , Li_2S_6 , Li_2S_4 and Li_2S) at varying potentials and elucidate their conversion pathways. Li_2S_4 and Li_2S_6 were predominantly observed, in which Li_2S_4 represents the key electrochemical intermediate dictating the overall SRR kinetics. Li_2S_6 , generated (consumed) through a comproportionation (disproportionation) reaction, does not directly participate in electrochemical reactions but significantly contributes to the polysulfide shuttling process. We found that the nitrogen, sulfur dual-doped holey graphene framework catalyst could help accelerate polysulfide conversion kinetics, leading to faster depletion of soluble lithium polysulfides at higher potential and hence mitigating the polysulfide shuttling effect and boosting output potential. These results highlight the electrocatalytic approach as a promising strategy for tackling the fundamental challenges regarding Li-S batteries.

The lithium sulfur (Li-S) battery represents an attractive, next-generation energy storage device because of its exceptional theoretical capacity of $1,672 \text{ mAh g}^{-1}$ and ultrahigh energy density of $2,600 \text{ Wh kg}^{-1}$ (refs. 7,8). Despite extensive efforts devoted to improving the practical performance of Li-S batteries^{3,9–11}, the fundamental reaction mechanism—particularly for the sulfur reduction reaction (SRR) during discharge—remains a topic of considerable debate^{4–6,12–15}. The SRR involves a complex multistep, 16-electron conversion from S_8 molecules to Li_2S solid, with multiple potential interwoven branches among a series of soluble lithium polysulfide (LiPS) intermediates. Soluble LiPS can readily shuttle across the cathode and anode, leading to rapid capacity fading. Recent investigations suggest that the conversion from high-order polysulfides to insoluble Li_2S_2/Li_2S represents the most difficult step, leading to an accumulation of soluble LiPS in the electrolyte and exacerbation of the shuttling issue¹⁶. An electrocatalytic process could help accelerate such conversion kinetics and reduce LiPS accumulation, and diverse electrocatalysts have shown promise in improving battery performance^{17–25}. However, the exact role of such electrocatalysts in modification of the SRR mechanism remains

elusive. A comprehensive elucidation of the SRR network and the electrocatalytic effect is essential for the rational design of electrocatalysts that can target specific steps to fundamentally solve the polysulfide shuttling problem.

Various approaches have been considered to elucidate the SRR mechanism, involving both experimental^{4,5,12–14,26–28} and computational efforts^{12,13,29–34}. Detailed mechanistic studies based on stand-alone electrochemistry are generally challenging, owing to the complex convolution of multiple electrochemical reactions and non-electrochemical side reactions (for example, LiPS disproportionations^{4,5,28}) at the same potential. Advanced in situ characterization, to identify and track different polysulfide species generated electrochemically or non-electrochemically, is essential for the interpretation and corroboration of electrochemical characteristics^{5,13,14,27}. Computational efforts have focused on predicting the energetics of different species and, furthermore, the reaction network^{12,34}. However, current studies do not treat the overall reaction network equilibrium and lack a description of potential-dependent properties such as polysulfide concentrations along the SRR process. This hinders direct comparison with

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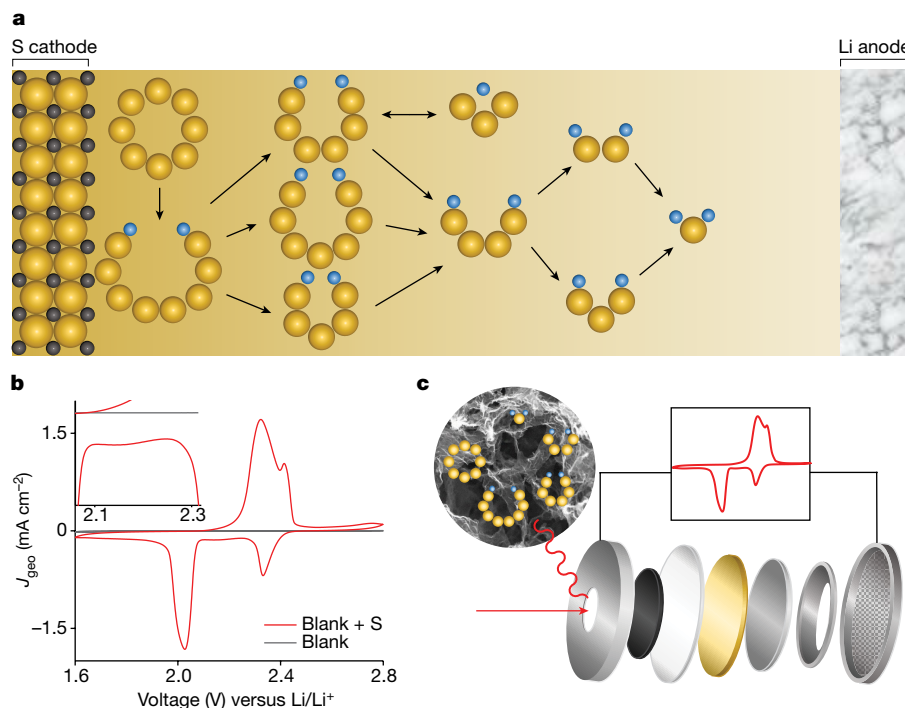


Fig. 1 | Polysulfide conversion reactions involved in the Li-S battery.

a, Schematic illustration of the SRR network involved in the Li-S battery. Blue and yellow spheres represent lithium and sulfur, respectively. SRR encompasses a sophisticated 16-electron conversion process from S_8 molecules to Li_2S solid, involving multiple soluble LiPS intermediates. **b**, CV of the SRR in a Li-S battery with N,S-HGF catalysts. The CV data were collected in a coin cell at a scanning rate of 0.05 mV s^{-1} (see Methods for details). The black baseline was obtained

using the same N,S-HGF in the blank electrolyte without sulfur, indicating the negligible double-layer capacity contribution to overall capacity. Inset shows the plateau in the approximate voltage range 2.11–2.25 V, which originated from the delayed conversion of Li_2S_8 to Li_2S_4 due to $Li_2S_8 + Li_2S_4 \rightleftharpoons 2 Li_2S_6$ comproportionation/disproportionation reactions. J_{geo} , geometric current density. **c**, Schematic illustration of the in situ Raman technique used in this study.

experimental results and therefore it is highly desirable to determine the detailed mechanism based on ab initio energetics with validation from both electrochemical and in situ techniques.

Here we report a systematic investigation of the electrocatalytic SRR mechanism. We chose the previously developed nitrogen, sulfur, dual-doped holey graphene framework (N,S-HGF) electrocatalysts and non-doped HGF as model systems to explore the impact of catalysts in modification of the reaction network and kinetics. Combining cyclic voltammetry (CV), in situ Raman spectroscopy and density functional theory (DFT) calculations, we establish (1) a detailed reaction network, (2) elucidate the dominant reaction pathway before and after the central Li_2S_4 intermediate, (3) identify the key species as S_8 , Li_2S_8 , Li_2S_6 , Li_2S_4 and Li_2S and (4) determine that the non-electrochemical comproportionation/disproportionation reaction between Li_2S_8 and Li_2S_4 is the main path for formation or consumption of Li_2S_6 . Comparison between N,S-HGF and HGF confirmed the same key species in the reaction network and the N,S-HGF catalyst accelerated LiPS conversion, leading to more rapid depletion of LiPSs at higher potential to mitigate the polysulfide shuttling effect and yield a higher output potential. These results emphasize electrocatalysis as a promising strategy to address fundamental Li-S battery challenges.

Reaction network in SRR and CV results

It has previously been shown that N,S-HGF can markedly accelerate SRR kinetics in comparison with non-doped HGF¹⁶. The CV curve for SRR with N,S-HGF (red curve in Fig. 1b) exhibited two main peaks during discharge: one appearing around 2.2–2.5 V and a second around 1.9–2.1 V. The charge number calculated from the integrated area in CV was converted to the formal electron transfer number in a full

$16e^-$ process (Supplementary Fig. 1 and Supplementary Table 1). By separation of the overall discharge process at the onset of the second peak (2.11 V), a charge transfer ratio of 4.08:11.92 (approximately equal to 1:3) was obtained, which was also validated by galvanostatic charge–discharge testing (Supplementary Fig. 2). This 1:3 charge transfer ratio suggests that Li_2S_4 is the primary intermediate separating these two reduction peaks, because the reaction $S_8 + 4Li^+ + 4e^- \rightarrow 2Li_2S_4$ involves 4 electrons out of the formally overall 16 electrons transferred, and the subsequent conversion $2Li_2S_4 + 12Li^+ + 12e^- \rightarrow 8Li_2S$ involves 12 electrons.

Interestingly, a non-zero plateau was observed in the voltage range 2.11–2.25 V between the two peaks (inset in Fig. 1b). Charge integration results, as represented in Fig. 2a by coloured regions, illustrate the non-negligible contribution of the plateau region between the two major redox peaks—about one out of 16 electrons per S_8 molecule in SRR. In comparison with a control group lacking sulfur (black baseline in Fig. 1b), the double-layer capacity contribution to this plateau has been eliminated. Considering the instability and complexity of polysulfides^{4,5,28}, the voltage ranges of electrochemical reactions reflected on CV can be affected by comproportionation/disproportionation reactions, although such non-electrochemical processes are not directly detectable via CV measurement.

To investigate the chemical origin of such a plateau we used first-principles calculations to explore the fundamental energetics among different polysulfide intermediates. Computational modelling of the SRR network began with the conversion from S_8 to Li_2S_8 , yielding a calculated output potential of 2.41 V, the highest among all steps. Further conversion of the Li_2S_8 molecule involved multiple possible branches whereas the calculated energetics show that the pathway forming two Li_2S_4 is the most exergonic and hence the favoured one, yielding an output potential of 2.24 V (Fig. 2d).

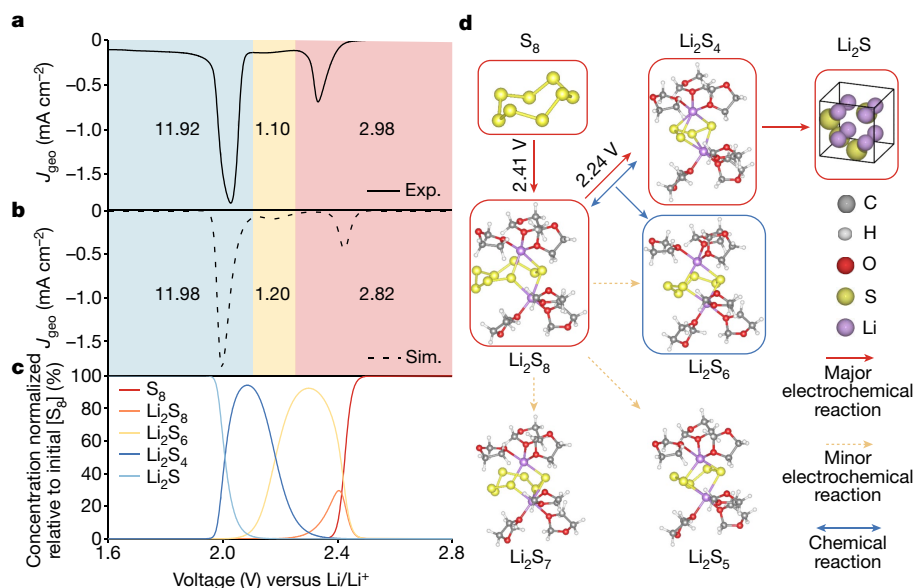


Fig. 2 | Charge analysis and reaction network for the SRR. a, Experimental CV curve (Exp.) with charge integration results, separated into three potential regions with a charge ratio of 2.98:1.10:11.92 from high to low potential. The CV data were collected in a coin cell at a scanning rate of 0.05 mV s⁻¹ (see Methods for details). **b**, Simulated CV curve (Sim.) from first principles with the charge integration results, separated into the same three regions as in **a**, with charge ratio 2.82:1.20:11.98. **c**, Simulated voltage-dependent concentrations of the major species considered: S₈, Li₂S₈, Li₂S₆, Li₂S₄ and Li₂S. Concentrations are normalized according to sulfur content. **d**, The dominant reaction mechanism

suggested by DFT energetics: S₈ → Li₂S₈ → 2Li₂S₄ → 8Li₂S (Li₂S₈ + Li₂S₄ ⇌ 2Li₂S₆), in which the chemical disproportionation part is shown in parentheses. Solid red and dotted yellow lines indicate major and minor electrochemical reactions, respectively, and blue lines indicate chemical reactions. Major products are indicated by red and blue boxes, corresponding to electrochemical and chemical origin, respectively. Thermodynamic output potentials are denoted for major electrochemical reactions. The catalytic site-dependent output potentials for Li₂S₄ → Li₂S are detailed in Fig. 5.

Because the experimentally observed plateau might have originated from the delayed electrochemical conversion of comproportionation or disproportionation products, we checked the possibilities starting with one Li₂S₈ molecule and one Li₂S₄ molecule, or two Li₂S₄ molecules, and found that the reaction Li₂S₈ + Li₂S₄ → 2Li₂S₆ was the only exergonic one, with a reaction Gibbs free energy of -0.16 eV, whereas the disproportionation reactions involving either Li₂S₈ and Li₂S₆ or Li₂S₄ and Li₂S₆ were found to be endergonic. These results suggest that Li₂S₆ formation by comproportionation of Li₂S₄ and Li₂S₈ is the only chemical elementary step that significantly competes with the electrochemical reaction network.

Combining the aforementioned DFT balances (Fig. 2d) with voltage effects, the potential-dependent concentrations of different polysulfides were simulated, giving a sequence of dominant LIPS species as S₈, Li₂S₈, Li₂S₆, Li₂S₄ and Li₂S with reducing potential (Fig. 2c). The simulated CV curve was further derived from the concentrations, giving a charge ratio of 2.82:1.20:11.98 in the red, yellow and blue zones, respectively, of Fig. 2b, which matches well with the experimental ratio shown in Fig. 2a.

Although Li₂S₆ appeared immediately after Li₂S₈, it was not produced by electrochemical reduction of Li₂S₈. Instead, a fraction of Li₂S₈ (about one-third) was electrochemically transformed into Li₂S₄, providing a fractional amount of charge in the red region shown in Fig. 2a,b (around two-thirds e), whereas the remainder underwent comproportionation with the produced Li₂S₄ to yield a large concentration of Li₂S₆ at roughly 2.25 V. Note that the exergonic nature of Li₂S₈ + Li₂S₄ → 2Li₂S₆ comproportionation provided additional driving force to initiate the electrochemical reduction of Li₂S₈ to Li₂S₄ at a potential higher (about 2.35 V) than its equilibrium (2.24 V). At lower potential, when Li₂S₈ is largely consumed, the comproportionation reaction operates backwards to disproportionate Li₂S₆ into Li₂S₄ and Li₂S₈ (2Li₂S₆ → Li₂S₈ + Li₂S₄) in which Li₂S₈ is electrochemically reduced to Li₂S₄, resulting in the discharge plateau seen in the yellow region. We found that the direct reduction

of Li₂S₆ to Li₂S₄, Li₂S₃ or other lower-order polysulfides cannot occur at a potential higher than 1.97 V, making the disproportionation reaction the only viable path in this potential regime (Supplementary Note 1). At even lower potential in the blue zone, Li₂S₄ is eventually reduced to Li₂S, involving the 12 extra electrons of the electrochemical reduction reaction.

In situ Raman study on SRR

We used in situ Raman spectroscopy (Supplementary Fig. 3) to probe the specific reaction intermediates along a discharge CV scan (Fig. 3a,b). Initially the typical S₈ peak at 469 cm⁻¹ confirmed the existence of elemental sulfur³⁰. With reducing potential, the S₈ signal gradually decreased and mostly disappeared at approximately 2.36 V, accompanied by the emergence of the Li₂S₈ signal at 508 cm⁻¹ starting at about 2.44 V, along with the appearance of the Li₂S₆ peak at 399 cm⁻¹. As discussed in the computation section, this occurs by electrochemical transformation of Li₂S₈ to Li₂S₄ and rapid comproportionation between the formed Li₂S₄ and remaining Li₂S₈ to yield Li₂S₆. The Li₂S₈ peak at 508 cm⁻¹ reached maximum at around 2.32 V, at which a deconvoluted peak at 501 cm⁻¹ for Li₂S₄ emerged. At 2.18 V, the 508 cm⁻¹ Li₂S₈ peak largely disappeared and the Li₂S₄ peak at 501 cm⁻¹ reached its maximum. As the potential decreased, Li₂S₄ became the main polysulfide species and Li₂S₆, formed through the comproportionation reaction (Li₂S₈ + Li₂S₄ ⇌ 2Li₂S₆), was also significantly present. The Li₂S₆ peak at 399 cm⁻¹ began to decrease at roughly 2.30 V and almost disappeared at 2.02 V. Similarly, most Li₂S₄ disappeared at around 2.00 V, indicating the conversion from Li₂S₄ to Li₂S_x (x = 1, 2, 3; Fig. 5). A small amount of Li₂S₄ remained at the low-voltage region below 1.90 V, probably due to incomplete conversion. Additionally, a minor Raman peak at 531 cm⁻¹ was assigned to LiS₃. Our calculations show that LiS₃ originated from an electrochemically passive branch (Li₂S₆ ⇌ 2LiS₃) and was present as a minor species (under 3% of Li₂S₆; Supplementary Fig. 4).

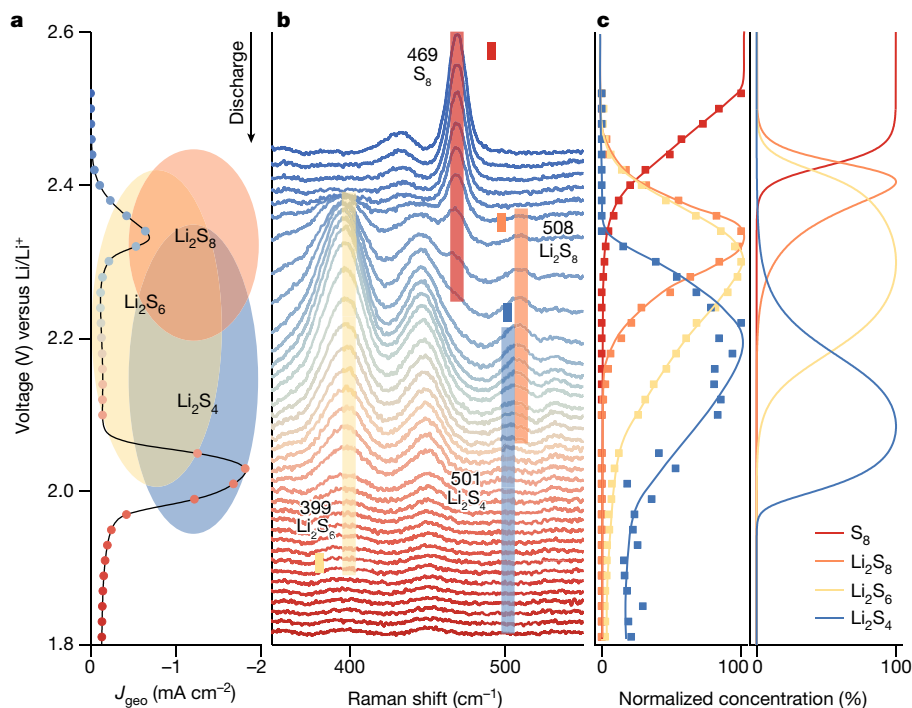


Fig. 3 | In situ Raman results during discharge with the N,S-HGF catalytic electrode. a, b, CV profile (a) and experimental in situ Raman spectra (b), with colours corresponding to voltages. The Raman cell was run with a discharge CV scan at 0.05 mV s^{-1} when data were being collected (see Methods for details). Characteristic peaks used to quantify intermediates are marked in corresponding colour shades. Small labels with darker colour indicate

computed frequency values. **c,** Comparison between voltage-dependent experimental concentrations (left) derived from in situ Raman spectra in **b** and simulated concentrations from DFT (right). The normalized concentration of each species is calculated by dividing the concentration at each voltage by the highest concentration of that particular species over the whole discharge process.

Li_2S_3 had negligible impact on the equilibrium of other species or the overall reaction network, and is thus omitted in our subsequent analysis.

The voltage-dependent concentration profile of each LiPS derived from the peak area (Fig. 3c, left) was further compared with the computational results (Fig. 3c, right). The comparison showed a similar sequence of concentration evolution for S_8 , Li_2S_8 , Li_2S_6 and Li_2S_4 with decreasing potential. Notably, Li_2S_6 appeared at a potential similar to that of Li_2S_8 and was depleted at a potential similar to that of Li_2S_4 , suggesting the dynamic balance among these three species through the comproportionation/disproportionation reactions.

We also note that the experimental peak for Li_2S_4 appeared at a slightly higher potential value (by 0.09 V) than the theoretical prediction. Only thermodynamics was considered in the simulated voltage-dependent equilibrium concentration whereas the formation of Li_2S_6 could have been slow because of kinetics and diffusion barriers, and thus Li_2S_4 accumulation may have started at a slightly higher voltage. This further validates the origin of Li_2S_6 from the comproportionation reaction. In addition, the low solubility of Li_2S_4 in the electrolyte (10 mM for Li_2S_4 versus 1 M for Li_2S_6 and 0.5 M for Li_2S_8 ; Supplementary Note 2) may also have contributed to an apparent shift in the Li_2S_4 profile: taking the solubility limit into account shifts the onset of the simulated Li_2S_4 peak position from about 2.25 V to roughly 2.35 V , and a better match with experimental results (Supplementary Fig. 5).

Together, these in situ Raman spectroscopy analyses provide semi-quantitative polysulfide tracking, in agreement with DFT computed values and thus robustly validating the SRR molecular pathway obtained by theory: $\text{S}_8 \rightarrow \text{Li}_2\text{S}_8 \rightarrow 2\text{Li}_2\text{S}_4$ ($\text{Li}_2\text{S}_8 + \text{Li}_2\text{S}_4 \rightleftharpoons 2\text{Li}_2\text{S}_6$) $\rightarrow 8\text{Li}_2\text{S}$. On the basis of Raman studies, we also estimated the relative concentration ratio of $1.7:5.5:0.4$ for the maximum concentration point of $\text{Li}_2\text{S}_8:\text{Li}_2\text{S}_6:\text{Li}_2\text{S}_4$ (Supplementary Table 2). Thus, both theoretical

and experimental studies suggest that Li_2S_6 represents the dominant species during discharge and is a major contributor to the shuttling effect.

The role of catalysis in the SRR network

To further understand the influence of electrocatalysts on the SRR network, the non-doped HGF was studied as a less active catalytic system for comparison with N,S-HGF (Supplementary Fig. 6). Overall, our in situ Raman spectra studies (Supplementary Fig. 7) show that HGF has a similar polysulfide evolution sequence, $\text{Li}_2\text{S}_8 \rightarrow \text{Li}_2\text{S}_6 \rightarrow \text{Li}_2\text{S}_4$, but with different voltage ranges for each species (Fig. 4).

The first step in discharge at high potential was weakly modified between the two different catalysts, the peak centre (Supplementary Table 3) for Li_2S_8 being slightly delayed in HGF (2.29 V) compared with N,S-HGF (2.32 V). Transformation of Li_2S_6 was more sensitive and markedly delayed in HGF, with an average peak value at 2.19 V (compared with 2.27 V in N,S-HGF) and with about 20% (relative to the peak concentration of Li_2S_6) remaining at 1.80 V in HGF (compared with less than 3% remaining at 1.80 V in N,S-HGF). A similar delay was observed for Li_2S_4 , from higher overpotential in the later steps, with an average peak value at 2.12 V (compared with 2.14 V in N,S-HGF) and with more than 30% remaining at 1.80 V in HGF (compared with 17% remaining in N,S-HGF). The delayed depletion of Li_2S_4 and Li_2S_6 until a much lower potential with non-doped HGF electrodes implies a more sluggish conversion kinetics to lower-order polysulfide, potentially worsening the shuttling effect.

Our calculations described above indicate that direct electrochemical reduction of Li_2S_6 to lower-order LiPSs above 1.97 V is unfavourable. Rather, elimination of Li_2S_6 relies on a thermodynamically unfavourable disproportionation reaction ($2\text{Li}_2\text{S}_6 \rightarrow \text{Li}_2\text{S}_8 + \text{Li}_2\text{S}_4$) that requires rapid

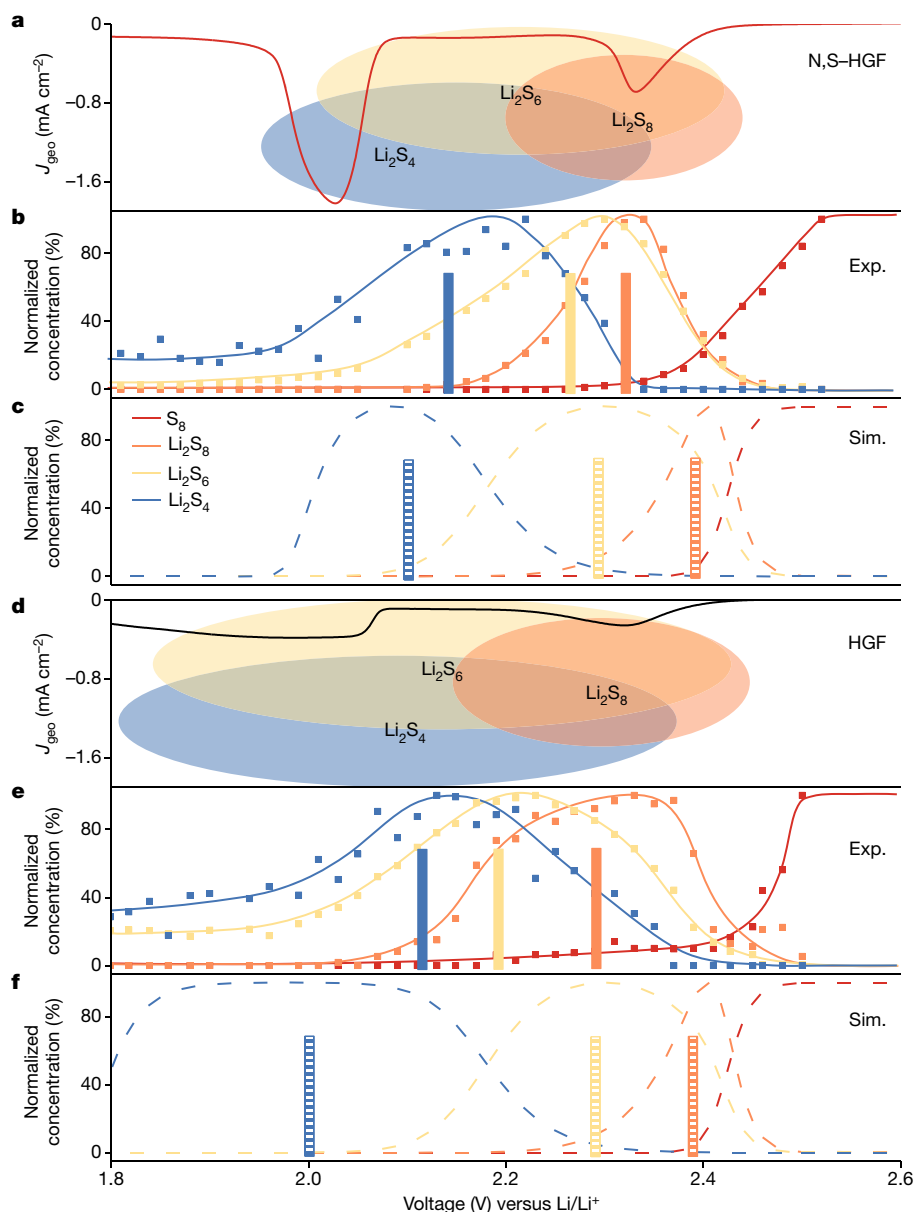


Fig. 4 | Comparison of different catalysts in SRR. **a,d**, Experimental CV curves for N,S-HGF (**a**) and HGF (**d**). **b,e**, Voltage-dependent concentration for each polysulfide species in N,S-HGF (**b**) and HGF (**e**) derived from experimental in situ Raman spectra. The Raman cell was run with a discharge CV scan at $0.05\ mV\ s^{-1}$ when Raman data were being collected (see Methods for details). **c,f**, Simulated voltage-dependent concentration for each polysulfide species in N,S-HGF (**c**) and HGF (**f**). The normalized concentration of each species is

calculated by dividing the concentration at each voltage by the highest concentration of that particular species over the whole discharge process.

b,c,e,f, Coloured bars represent the peak centres of voltage-dependent concentration curves (Supplementary Table 3) for each species: orange for Li_2S_8 , yellow for Li_2S_6 and blue for Li_2S_4 . Solid lines and bars in **b,e** represent the experimental results, and dashed lines and bars in **c,f** represent the simulated results.

depletion of Li_2S_4 to proceed. In this case, a slower conversion kinetics of Li_2S_4 could seriously delay the consumption of Li_2S_6 until a much lower potential regime (under 1.97 V) in which direct electrochemical reduction of Li_2S_6 may also start to occur. The conversion of Li_2S_4 thus dominates electrocatalytic performance, especially at the start of the second stage of SRR, which is consistent with the previously reported critical role of Li_2S_4 in SRR overpotential²⁰.

Overall, the slower conversion kinetics of Li_2S_6 resulted in its wider accumulation potential range, which exacerbated the polysulfide shuttling problem. This is also reflected by the reduced charge number at high potential and within the overall SRR process: 2.36 and 9.24 electrons per S_8 molecule in the HGF system compared with 3.61 and 14.17 in the N,S-HGF system (Supplementary Table 1). These CV and

in situ Raman studies showed distinct SRR kinetics between HGF and N,S-HGF, highlighting the fundamental benefits yielded by efficient catalysts.

To understand the distinct potential ranges for these two electrocatalysts, we further investigated the reaction pathways of the second stage—that is, the conversion from Li_2S_4 to Li_2S . Considering all possible $2e^-$, $4e^-$ and $6e^-$ steps starting from Li_2S_4 (Fig. 5), we examined a total of 12 different reaction pathways in the presence of various catalyst sites: armchair edge of graphene, zigzag edge and inner defects in the graphene plane with various doping situations (non-doped, S-doped, N-doped and N,S-doped; Supplementary Note 3). Two pathways were found to yield the highest output potential: (1) one $4e^-$ step: $Li_2S_4 + 4Li^+ + 4e^- \rightarrow Li_2S_2 + 2Li_2S$, followed by one $2e^-$ step: $Li_2S_2 + 2Li^+ + 2e^- \rightarrow 2Li_2S$

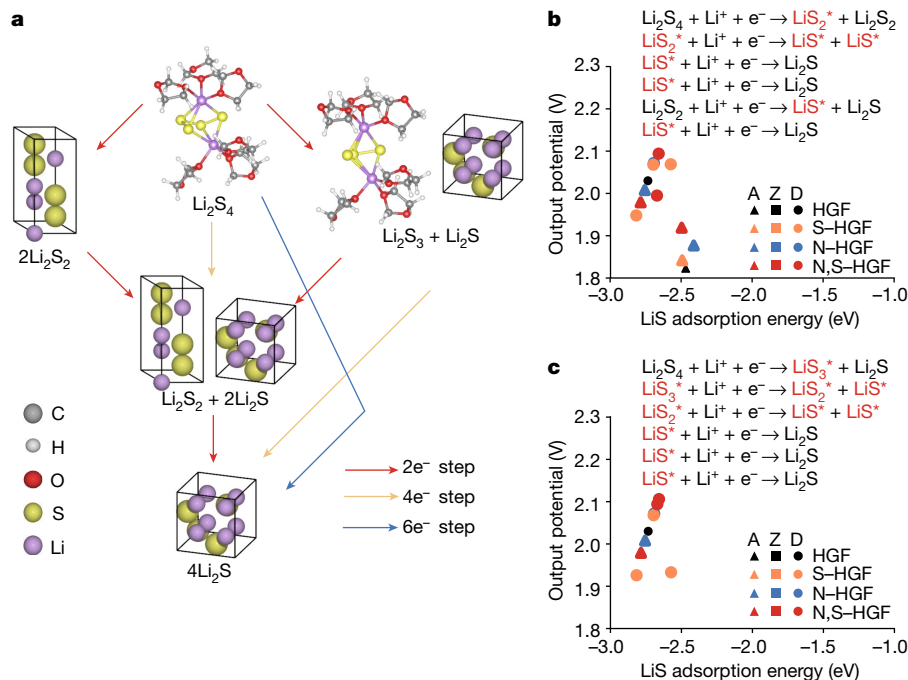


Fig. 5 | Simulated site-specific output potential of Li₂S₄ → Li₂S conversion. **a**, Different potential combinations of 2e⁻, 4e⁻ and 6e⁻ steps considered for the second stage of SRR, the conversion of Li₂S₄ to Li₂S. Red, yellow and blue indicate 2e⁻, 4e⁻ and 6e⁻ steps, respectively. **b,c**, Simulated multistep output potential from Li₂S₄ to Li₂S for the two pathways with highest output potentials

(Fig. 5b); and (2) a pathway consisting of one 6e⁻ step—that is, at least one intermediate species is adsorbed on the surface during the reduction process: Li₂S₄ + 6Li⁺ + 6e⁻ → 4Li₂S (Fig. 5c). The results clearly show that, in both pathways, N,S-codoped sites exhibited higher output potential, 2.11 V compared with 2.03 V for non-doped sites, aligning with experimental results and showcasing N,S-HGF's superior performance¹⁶.

Interestingly, the inner defect sites (Supplementary Figs. 8 and 9) appear closer to the top of the output potential plot (circles in Fig. 5b,c) compared with armchair and zigzag edge sites (triangles and squares, respectively), confirming the effectiveness of defect engineering, together with heteroatom doping, in holey graphene for electrocatalysis. It is worth mentioning that conversion from LiS to Li₂S solid is the potential limiting step for most of the sites with relatively large output potentials (approximately 2.1 V), with one site having the adsorption of LiS as the potential limiting step. This indicates that the final conversion from Li₂S₂ to Li₂S is the potential limiting step, consistent with the experimental observation in which the final steps exhibited the largest overpotential¹⁶. Therefore, LiS adsorption energy can be used as a descriptor to classify site output potentials. Moreover, a smaller output potential in the second stage has far-reaching effects: the sluggish conversion of Li₂S₄ to lower-order polysulfides could markedly retard the thermodynamically unfavourable disproportionation reaction (2Li₂S₆ → Li₂S₈ + Li₂S₄), the essential path for consumption of Li₂S₆.

Consistent with the experimental results, the simulated potential-dependent concentrations for the HGF electrode show more sluggish conversion: lower depletion potentials for Li₂S₄ and Li₂S₆ species (1.85 and 2.00 V for HGF versus 2.00 and 2.05 V for N,S-HGF; Fig. 4). The simulation of HGF and N,S-HGF effective output potential differs significantly only in the second stage, largely comparable to experimental results. Such a close correlation between experiment and theory further validates the electrocatalytic strategy for improved Li-S

batteries: an accelerated polysulfide conversion kinetics can not only produce a larger output potential, but also narrow the potential range in which LiPSs could appear and effectively mitigate the polysulfide shuttling effect.

Conclusion

In conclusion, a combined experimental and theoretical investigation has allowed us to decipher and establish the complex reaction network for the 16-electron SRR, showing two stages separated by the central Li₂S₄ intermediate. Our studies indicate that Li₂S₄ and Li₂S₆ represent the dominant intermediates, in which Li₂S₄ is the key electrochemical intermediate controlling the overall SRR kinetics—particularly in the more sluggish second reduction stage; Li₂S₆ is generated by the comproportionation reaction between Li₂S₈ and Li₂S₄, does not directly participate in electrochemical reactions, but contributes substantially to the shuttling effect because of its high solubility and energetically favourable accumulation in the electrolyte. It is found that the optimized N,S-HGF electrocatalytic electrode markedly accelerates the conversion of high-order LiPSs, leading to more rapid depletion of soluble LiPSs at a higher potential regime and hence mitigating the polysulfide shuttling effect and boosting output potential. This study resolves the fundamental reaction network in SRR and offers valuable insights into electrocatalyst design for improved Li-S batteries. For example, considering the origin of Li₂S₆ (Li₂S₈ + Li₂S₄ → 2Li₂S₆), designing electrocatalysts with enhanced Li₂S₈ or Li₂S₄ adsorption could restrain these species on the catalyst surface, thereby suppressing comproportionation reactions and limiting Li₂S₆ formation. Furthermore, this methodology can be applied to understanding the sulfur evolution reaction to guide the design of bifunctional sulfur catalysts for acceleration of both SRR and sulfur evolution reaction processes, which is essential for the development of robust Li-S batteries.

Online content

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Methods

Synthesis of graphene oxide and heteroatom-doped holey graphene framework

The heteroatom-doped HGF can function as a conductive scaffold with flexible pore structure and exhibit tunable electrocatalytic properties in SRR^{16,17}, and hence serves as a good model system for investigation of the fundamental reaction mechanism. Furthermore, compared with metal oxides or sulfides that are usually in the form of nanoparticles and that require additional conductive additives and binders, the free-standing HGF electrode could offer a more robust and simplified binder-free model system for systematic investigation of the SRR process. The specific heteroatom-doped N,S-HGF was chosen for its superior SRR kinetics, markedly low activation energy and excellent performance in Li-S batteries¹⁶, and non-doped HGF was selected as a control for comparison to evaluate the role of the catalyst in modification of the kinetics along the complex SRR network.

A typical hydrothermal process was used to construct the HGF structure, with interconnected micro- and nanopores. Graphene oxide was prepared using a modified Hummers' method³⁵. Briefly, 6 g of natural graphite (325 mesh, Sigma-Aldrich) was added to 140 ml of concentrated sulfuric acid under vigorous stirring in an ice-water bath followed by the slow addition of 3 g of sodium nitrate (Sigma-Aldrich) and 18 g of potassium permanganate (Sigma-Aldrich). Owing to the strong acidity of sulfuric acid and highly oxidative nature of sodium nitrate and potassium permanganate, it is necessary to keep the temperature near 0 °C to slow the oxidation process and avoid potential safety concerns. Following stirring for 30 min, the reaction system was transferred to a water bath at about 50 °C under continuous stirring until the mixture had formed a thick paste. The system was then transferred back to the ice-water bath followed by the dropwise addition of around 1 l of iced deionized water. The mixture was then centrifuged and washed three times with 1:10 HCl aqueous solution followed by repeated washing with deionized water. The final solution was dialysed for 1 week to remove extra H⁺ ions absorbed on graphene oxide surfaces.

Heteroatom-doped HGFs were synthesized by reacting the dopant sources and H₂O₂ with graphene oxide aqueous dispersion through a typical hydrothermal method. N,S-HGF was synthesized by mixing 30 ml of 2 mg ml⁻¹ graphene oxide aqueous dispersion solution, 45 µl of 30% H₂O₂ aqueous solution and 3.0 g of NH₄SCN, followed by sonication and hydrothermal reaction at 180 °C for 6 h in an autoclave, to produce a free-standing N,S-HGF hydrogel. The hydrogel was then freeze-dried and annealed at 900 °C in Ar for 1 h to obtain the N,S-HGF aerogel. The control sample HGF was synthesized by substitution of dopant sources with ascorbic acid, following the same procedures.

Preparation of electrolyte and Li₂S_x solutions

The electrolyte (denoted as blank electrolyte) comprised 1 M lithium bis(trifluoromethanesulfonyl)imide (Sigma-Aldrich) and 0.2 M lithium nitrate (Sigma-Aldrich) in the mixed 1,2-dimethoxyethane (Sigma-Aldrich) and 1,3-dioxolane (Sigma-Aldrich) solution (1:1 by volume). The Li₂S₆ catholyte (0.1 M) was prepared by reacting the sublimed sulfur (Sigma-Aldrich) with Li₂S (Sigma-Aldrich) in stoichiometric proportion in the blank electrolyte. The mixture was vigorously stirred at 50 °C in an Ar-filled glovebox overnight to produce a brownish-red Li₂S₆ catholyte solution.

Electrochemical measurements

The electrochemical performance of the catalyst was conducted in CR2032 coin cells assembled in an Ar-filled glovebox. The catalyst electrode was prepared by direct pressing of the aerogel into a free-standing thin film. Next, the catholyte (0.1 M Li₂S₆ in the blank electrolyte) was used directly as the sulfur source for drop-casting in the catalyst electrode. In our experiment we specifically used a low mass ratio of sulfur (33%) in the cathodes to ensure complete conversion from S₈ to Li₂S for

mechanistic understanding. The sulfur cathodes were then directly assembled into a CR2032 coin cell with lithium foil, Celgard 2500 separator and blank electrolyte.

To better probe the different stages in the SRR we performed CV measurements followed by quantitative charge analysis, by integration of the peak area in different potential regions. CV curves were recorded in the voltage range 1.6–2.8 V at a scanning rate of 0.05 mV s⁻¹. The charge number transferred (*Q*) for the electrochemical steps in the SRR can be calculated from CV curves. *Q* is calculated by integrating the area enclosed in CV:

$$Q = \int_{t_1}^{t_2} i(t) dt = \frac{1}{\nu} \int_{V_1}^{V_2} i(V) dV = \frac{A}{\nu} \int_{V_1}^{V_2} j(V) dV = \frac{AS}{\nu} \\ dt = \frac{dV}{\nu}$$

where *Q* is charge (C), *i* is current intensity (A), *j* is current density (A cm⁻²), *A* is the geometric area of the electrode (cm²), *ν* is scan rate (V s⁻¹), *t* is time (s) and *S* is integrated area.

In situ Raman spectroscopy

Given the specific Raman activity of elemental sulfur and polysulfides, in situ Raman spectroscopy offers an attractive technique for identification and tracking of the polysulfide conversion process^{30,36–41}. Raman spectroscopy measurements were collected using a liquid nitrogen-cooled, charge-coupled device array detector with a Horiba Jobin Yvon T64000 open-frame confocal microscope using the ×10 objective, followed by a triple monochromator resulting in high spectral resolution (down to 0.15 cm⁻¹). The sample was subject to a 514 nm laser for 5 s and averaged 70 times. Data were collected with a 1,800 cm⁻¹ grating and 500 µm slit. For in situ Raman spectroscopy, a regular coin cell was modified with a transparent window on the cathode side to allow ingress of the laser. The same procedure as described for electrochemical measurements was used to assemble the Raman cell. The laser was focused on the electrolyte near the boundary between HGF and electrolyte (Supplementary Fig. 3). The Raman cell was run with a discharge CV scan at 0.05 mV s⁻¹ when data were being collected.

The resulting Raman spectra (Supplementary Fig. 10) were carefully corrected and deconvolved. Because the blank electrolyte shows multiple peaks in our spectral range of interest, in situ data were first corrected by 'blank' subtraction. Specifically, a blank spectrum was collected using the same cell set-up but without active sulfur being added and subtracted from the in situ data, to eliminate the influence from electrolyte peaks. Subsequently the in situ data were corrected by subtracting the baseline. The blank/baseline subtraction was conducted systematically using adaptive iteratively reweighted penalized least squares⁴² fitting implemented in the pybaseline package. The corrected Raman spectra were then loaded in *LabSpec* software to conduct peak deconvolution and assignments for different polysulfides. We note that peak assignment for different polysulfides is complicated and divergent in various studies^{30,36,39,43}, owing to the instability of various polysulfides and absence of pure polysulfide standard samples. Peaks used in our analysis were selected based on their behaviour during the SRR process, previous reports^{30,36,39,43} and our DFT simulated results. We deconvoluted all Raman peaks and analysed multiple peaks for each given LiPS to confirm the validity of the assignments (Supplementary Figs. 11 and 12), and primarily used the peaks at 501, 399, 508 and 469 cm⁻¹ for quantification of Li₂S₄, Li₂S₆, Li₂S₈ and S₈, respectively.

DFT calculations

Calculations were performed with DFT^{44,45} using the Vienna Ab initio Simulation Package⁴⁶. The strongly constrained and appropriately normed functional⁴⁷ was used, and further details are provided in

Supplementary Note 4 and Supplementary Table 4. Benchmarking and comparison between different functionals are provided in Supplementary Note 5 and Supplementary Tables 5 and 6. Solvation effects are described using a microsolvation model: the first solvation shell is described using explicit 1,3-dioxolane molecules and the remainder by an implicit dielectric model as implemented in the VASPsol⁴⁸ add-on package (microsolvent structures are given in Supplementary Fig. 13 and Supplementary Tables 7 and 8). We note that the choice of the explicit solvent molecule is an approximation, and we also performed calculations with explicit 1,2-dimethoxyethane solvation for Li₂S₄, which showed the same Li-O coordination number with very similar energies (Supplementary Note 6 and Supplementary Table 9). Details of Gibbs free energies of polysulfide species are provided in Supplementary Note 7 and Supplementary Tables 10–15, and reaction Gibbs free energies in Supplementary Note 1. Details of potential-dependent concentration calculations and simulated CV curves are provided in Supplementary Note 8. The possible effects of solubility on the simulated potential-dependent concentration profile are discussed in Supplementary Note 2. Adsorption models (Supplementary Figs. 8, 9 and 14) are discussed in Supplementary Note 3. Volcano plot details (Supplementary Fig. 15) are discussed in Supplementary Note 9. Raman intensity (Supplementary Fig. 16) was calculated within the double-harmonic approximation, and details are provided in Supplementary Note 10 and Supplementary Table 16. A sensitivity analysis is provided in Supplementary Note 11 and Supplementary Fig. 17, showing that theoretical conclusions are unaffected by a reasonable change of calculated energies (up to 0.15 eV) arising from the considered approximations.

Data availability

The data that support the findings of this study are available in the main text, figures and Supplementary Information files. Source data are provided with this paper. All relevant data are available from the corresponding authors on request.

Code availability

The code used for simulated voltage-dependent concentrations and CV curves is available at https://github.com/lophocinalis/concentration_cv.

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Author contributions X.D. conceived the research. X.D. and R.L. designed the experimental research. P.S. and Z. Wei designed and performed density functional theory (DFT) calculations. R.L. performed experiments and conducted data analysis, with contributions from Z. Wei, L.P., L.Z., P.W., C.W., D.Z., H.L., Z. Wang, S.T., B.D., Y.H., P.S. and X.D. A.Z. and R.S. contributed to Raman data collection. R.L. and Z. Wei wrote the original draft. X.D. and P.S. revised the manuscript. All authors discussed the results and commented on the manuscript.

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Additional information

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